

Quantum Computing

Part 6 - Error Correction & Logical Qubits

Compiled by the Spaced Research Ledger

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A quantum computer's parts are unreliable in a way ordinary computer parts are not. Error correction is the engineering answer, and it works by spreading one unit of protected information across many unreliable physical ones. The protected unit is called a logical qubit, and the number of physical qubits it costs is the field's central price.

Two years ago the open question was whether this worked at all. That question is now settled, and the field has moved on to harder ones.

Section 1 covers surface codes, the leading scheme and the one that proved the principle. Section 2 covers the newer high-rate codes that cut the price by an order of magnitude. Section 3 covers cat qubits, which attack the problem from the hardware side instead. Section 4 covers magic states, the missing ingredient for useful computation.

Section 5 covers decoding, the classical computing problem hiding inside the quantum one. Section 6 covers the milestones themselves.

1. Surface Codes

The surface code arranges physical qubits in a flat grid and repeatedly measures their neighbours to detect errors without disturbing the stored information. Its code distance is the width of that grid, and a larger distance corrects more errors. It is the most thoroughly demonstrated scheme in the field, and the one that first crossed the threshold where error correction starts helping rather than hurting.

Recent Developments

The most recent work has moved from storing a protected qubit to operating on one. A team realized the full set of surface-code logical operations on a 107-qubit superconducting processor, including merge and split operations, a CNOT gate, a Hadamard gate, and phase gates [1]. The work used distance-3 rotated surface-code patches, multi-round syndrome extraction, and a neural-network decoder. Storage was the earlier milestone. Computation on stored logical qubits is the harder one, and this is a direct demonstration of it.

Google's landmark below-threshold result also received two refinements. The underlying paper was issued an author correction in April 2026 [2]. Its real-time decoder now reports an average latency of 63 microseconds at distance 5, sustained across up to a million cycles.

Current State

The result that settled the field's central question came from Google's Willow processor. It demonstrated the first quantum memory operating below the surface-code threshold, running a distance-7 code and a

distance-5 code across chip configurations of 105 and 72 physical qubits [2]. The distance-7 memory used 101 physical qubits and reached a logical error rate per correction cycle of $0.143\% \pm 0.003\%$ [3].

The number that matters most is not the error rate but how it changed with size. Moving from distance 5 to distance 7 suppressed the logical error rate by a factor of $\Lambda = 2.14 \pm 0.02$ [2]. That ratio is called lambda, and a lambda above 1 is the definition of operating below threshold. It means adding more physical qubits makes the logical qubit better rather than worse. Every scaling argument in the field rests on that being true.

Two further properties of the same experiment matter for practical machines. The distance-7 memory outlived its best individual physical qubit by a factor of 2.4 ± 0.3 , which is the break-even point [2]. It also held below-threshold performance while decoding in real time, not only when analyzing results afterward [2]. The work was peer-reviewed and published in Nature in 2025, having first appeared as a preprint in August 2024 [4].

Google has since shown that the surface code's layout is not fixed.

Dynamic surface codes on Willow alternate between different tilings using hexagonal, walking, and iSWAP circuits [5]. They reached an error suppression factor of 2.15 from distance 3 to distance 5, using only three couplers per qubit. They also cut leakage-induced correlations by more than a factor of ten. The work was published in Nature Physics, which gives the hexagonal-lattice scheme peer-reviewed standing [6].

It also allows dropout avoidance, meaning the code can route around qubits that have failed. Fixed schemes cannot do that.

2. High-Rate Codes

A quantum low-density parity-check code, usually shortened to qLDPC, protects information using far fewer physical qubits than a surface code. It does this by letting its error checks connect distant qubits rather than only neighbours. The cost is harder wiring. The benefit is a much better ratio of physical qubits to logical ones, which is the single largest term in what a useful machine will cost to build.

Recent Developments

The efficiency claims made in this area during 2026 are large. IQM reported that its directional tile qLDPC codes cut the per-logical, per-round error rate by a factor of 1,000 against surface codes, running on its existing Crystal hardware [7].

A trapped-ion demonstration reached break-even with a qLDPC code, encoding 4 logical qubits into 18 physical qubits [8]. Its logical error rates ran up to 9 times better than earlier superconducting demonstrations, and its logical qubits matched or outlived the physical qubits underneath them.

The most striking number came from a QuEra, Harvard, and MIT preprint. It reported a code encoding 1,156 logical qubits into 2,304 physical qubits, a ratio of roughly two to one [9]. The simulated error rate per logical qubit per round was near 1.3×10^{-13} , at a physical error rate of 0.1%. That range is sometimes called the teraquop regime.

Two qualifications belong with it. The result is a simulation establishing a quantum-memory baseline, not a hardware demonstration. It also does not yet cover the fault-tolerant logical gates that universal computation requires.

Current State

IBM's bivariate bicycle code, known as the gross code, is the established reference point. It encodes 12 logical qubits using 288 physical qubits, split evenly between 144 data qubits and 144 check qubits, for a ratio of 24 to 1 [10]. A standard surface code would need close to 3,000 physical qubits for equivalent protection, roughly ten times as many [10]. The saving comes from letting error checks connect qubits that are far apart, using long-range couplers to imitate three-dimensional connectivity on a flat chip [11].

IBM has also addressed the classical side of the problem. It built a decoder for qLDPC codes that runs in real time on ordinary classical hardware, without needing a high-performance computing system sitting next to the quantum machine [12].

3. Cat Qubits and Bosonic Encodings

A cat qubit stores information in an oscillator rather than a two-level system. It shapes that oscillator so one of the two error types becomes physically rare rather than merely corrected. Bit-flip errors are suppressed by the hardware itself, which hands a simpler correction problem to the code above it. The approach trades generality for a large head start on one specific failure mode.

Recent Developments

Alice & Bob reported bit-flip lifetimes of 33 to more than 60 minutes on its Galvanic Cat architecture, the design used in its Helium 2 chip [13]. That range overlaps an earlier figure from the same company rather than clearly replacing it, and Section 3's conflict note below covers the disagreement.

Current State

The proof that the approach scales came from AWS. Its Ocelot chip, published in Nature in February 2025, is the first integrated multi-cat-qubit chip to combine cat-qubit stabilization with repetition-code error correction on a scalable superconducting platform [14]. AWS reports bit-flip times on Ocelot approaching one second, at a logical error rate near 1.65% at distance 5 [14]. A standard transmon qubit's lifetime is 50 to 100 microseconds. The bit-flip figure is roughly four orders of magnitude better than the conventional device it replaces.

Alice & Bob has pushed the same idea further with squeezed cat qubits. The approach applies a parametric squeezing drive that narrows the cat state's distribution in phase space. It reported a 160-fold improvement in bit-flip lifetime over earlier cat qubits on the same chip [15]. The company separately reported bit-flip stability exceeding one hour on its Galvanic Cat architecture in September 2025, against roughly 7 minutes on its earlier Boson 4 chip [14].

Unresolved Conflicts

Alice & Bob's own reported bit-flip lifetime for the Galvanic Cat architecture is inconsistent across sources. One report puts it above one hour, measured against roughly 7 minutes on the company's previous chip [14]. Another gives a range of 33 to more than 60 minutes for the same architecture [13]. The two overlap rather than one clearly superseding the other, so both stand on record. A dated measurement from the company naming the specific chip and test conditions would settle it.

4. Magic States

Error-corrected qubits can only perform a restricted set of operations cheaply, and that set is not enough for general computation. The missing operations run on specially prepared states called magic states, which are consumed in the process. Producing them at sufficient quality is expected to dominate the running cost of a fault-tolerant machine, which is why the production techniques below matter as much as the codes themselves.

Recent Developments

Silicon Quantum Co, working with the Centre of Excellence for Quantum Computation, announced a magic-state cultivation technique it calls Magic Scroll [16]. It reports magic-state fidelities with error rates as low as 10^{-9} , using colour and bilayer codes in 6.6.6 and 4.8.8 configurations on two-qubit registers. The method exploits biased noise to reduce hook errors.

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The field's first experimental demonstration of magic-state distillation performed entirely on logical qubits came from QuEra, Harvard, and MIT, using a neutral-atom machine and published in Nature in 2025 [17]. The experiment used a dynamically reconfigurable architecture to encode and operate on many logical qubits in parallel during distillation [17]. In one configuration only 8 physical qubits in an error-detecting circuit were needed to distill a magic state of acceptable fidelity [18]. The team projected that roughly 40 qubits would push the fidelity further.

Two other demonstrations fill in the picture. A trapped-ion experiment produced logical magic states through code switching, using up to 15 qubits [18]. Its logical gate error rate beat the physical error rate underneath it. The work was published in Physical Review X in 2025. Rosenfeld and colleagues demonstrated magic-state cultivation on a superconducting processor in a December 2025 preprint, putting into hardware a low-overhead protocol that had existed only on paper [19].

5. Real-Time Decoding

Every round of error correction produces measurement data that a classical computer must interpret before the quantum machine can act on it. That interpretation is called decoding, and it has to finish faster than errors accumulate. A decoder that is accurate but slow is useless, which makes this a classical engineering problem sitting in the middle of a quantum one.

Recent Developments

The most consequential recent result is about what decoding no longer requires. IonQ demonstrated a real-time decoding pipeline running on a single Apple M4 Max processor, handling workloads of up to 408 logical qubits and more than a million T gates [20]. It added under 0.3% computational stretch at a two-qubit gate error rate of 10^{-4} . The significance is the absence of special hardware. A laptop-class processor kept pace with a machine at a scale the field has been treating as a hardware problem.

Custom hardware still sets the latency record. An FPGA-based neural-network decoder built into a superconducting processor performed real-time distance-3 surface-code correction with a deterministic closed-loop latency of 550 nanoseconds, of which 124 nanoseconds was decoding [21]. Its accuracy matched offline minimum-weight perfect-matching decoding, the usual quality benchmark.

Google approached the problem from the control side instead. It published a reinforcement-learning scheme in Nature in July 2026 that unifies calibration and computation on Willow [22]. The scheme reports a 3.5-fold improvement in logical stability and a 20% cut in logical error rate.

Current State

Decoding integrated directly into processor control systems is now established. A June 2026 Nature Communications paper demonstrated real-time, low-latency correction using a scalable FPGA decoder built into a superconducting processor's control system [23].

Riverlane has taken the hardware decoder furthest toward deployment. Its Local Clustering Decoder was published in Nature Communications in December 2025 as a hardware decoder delivering real-time, scalable surface-code correction [24]. Implemented on FPGA hardware, it decodes a round in under a microsecond while staying accurate and adapting to changing noise [25]. It is described as the first hardware decoder to combine all three properties at once.

The decoder is deployed rather than demonstrated. Riverlane's Deltaflow stack runs across Infleqtion, Oxford Quantum Circuits, Rigetti Computing, and Oak Ridge National Laboratory, spanning superconducting, spin, trapped-ion, and neutral-atom platforms [25]. Deltaflow 2 went into a UK commercial data centre in July 2025, co-located with Oxford Quantum Circuits' machine [26]. It introduced streaming quantum memory, which protects information continuously during a computation rather than only between correction rounds.

6. Below-Threshold and Break-Even Milestones

This section collects the results whose importance is the milestone itself rather than the method. Break-even means a protected qubit outlives the physical qubits it is built from. Below threshold means adding more physical qubits improves rather than degrades the logical one. Both are thresholds in the ordinary sense, and crossing them changes what is worth attempting next.

Recent Developments

A joint team from the University of Innsbruck, RWTH Aachen, Forschungszentrum Jülich, and AQT ran a complete fault-tolerant algorithm end to end [27]. The algorithm was Grover's search on three logical qubits

encoded across eight physical qubits, executed on a trapped-ion processor. It ran without mid-circuit measurements, using feed-forward control instead. Previous fault-tolerant demonstrations have generally been components rather than whole algorithms.

Quantinuum reported the largest count of protected qubits to date. Its 98-physical-qubit Helios processor ran computations using up to 94 error-detected logical qubits and 48 error-corrected ones, at a two-to-one encoding ratio achieved through code concatenation [28]. State preparation and measurement fidelity was 99.99%. The company describes the result as beyond break-even, with logical error rates at least ten times lower than the matching physical-qubit operations.

Current State

A 2026 experiment reached below-threshold correction on a distance-7 surface code using all-microwave leakage suppression [29]. It reported an error suppression factor of $\Lambda = 1.40(6)$, and cut average leakage population by a factor of 72 after 40 correction cycles. Leakage is the failure mode where a qubit drifts out of its two working states entirely, and ordinary error correction does not address it.

About This Report

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